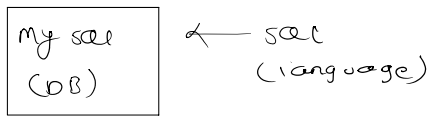


* Database : MySQL — Database
SQL — is a language use to talk with database



* What is database

→ It is a collection of data in a format that can be easily accessed.

Why database?

- can store a large data
- features like security, scalability etc
- Easier to insert, update, delete data.

* Types of databases — Here we learn about two type of database

SQL / Relational Relational database (data store in table)	NO-SQL / Non-Relational Non-relational database (data stored in document key-value pairs etc)
<u>e.g</u> MySQL, Oracle PostgreSQL etc	<u>e.g</u> MongoDB, Cassandra Neo4j etc.

* SQL (Structured Query Language)

SQL is a programming language used to interact with relational databases.

* Table in SQL

Schema: - The structure of information we stored in a database is called schema.

id	name	email	followers	following
1	adam	@gmail.com	200	50
2	steve	@gmail.com	110	100

id
name
email
followers
following

Schema

* rows — A row information about one entity

Note: rows also known as a tuple

e.g

id	name	email	followers
1	adam	@gmail	100

— single row

id	name	email	follower
1	adam	@gmail	100

— single row

* column :- column define type of information stored in a row

e.x id - student id
name - student name

* 1st database :-

create database college; — To create database

drop database college; — To delete database.

Note :- for creating table in database we need to use first

e.x use college — use database for creating table

① create Table student (
column_name datatype, constraints — syntax
Roll-no INT,
name varchar(20)
email varchar(20)
);

* Database queries

Database Queries

CREATE DATABASE db_name;

CREATE DATABASE IF NOT EXISTS db_name;

DROP DATABASE db_name;

DROP DATABASE IF EXISTS db_name;

SHOW DATABASES;

SHOW TABLES;

Data Types

DATATYPE	DESCRIPTION	USAGE
CHAR	string(0-255), can store characters of fixed length	CHAR(50)
VARCHAR	string(0-255), can store characters up to given length	VARCHAR(50)
BLOB	string(0-65535), can store binary large object	BLOB(1000)
INT	integer(-2,147,483,648 to 2,147,483,647)	INT
TINYINT	integer(-128 to 127)	TINYINT
BIGINT	integer(-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807)	BIGINT
BIT	can store x-bit values, x can range from 1 to 64	BIT(2)
FLOAT	Decimal number - with precision to 23 digits	FLOAT
DOUBLE	Decimal number - with 24 to 53 digits	DOUBLE
BOOLEAN	Boolean values 0 or 1	BOOLEAN
DATE	date in format of YYYY-MM-DD ranging from 1000-01-01 to 9999-12-31	DATE
YEAR	year in 4 digits format ranging from 1901 to 2155	YEAR